BLOG POST

Good Views Hidden in Plain Sight

A Closer Look at the Next-Token Baselines in "Proving and Improving the Platonic Representation Hypothesis with JEPAs"

Thomas Bernwald

4 November 2026 · preliminary, single-seed experiments · blog post
with interactive figures at thomasaas.github.io/good-views.html

9 November 2026: added readouts that average several sampled continuations, which bring NTP close to the JEPA at web scale, and Appendix A5.

7 November 2026: added further results and updated the text for easier readability.

AI Usage



TL;DR. Balestriero and LeCun's *Proving and Improving the Platonic Representation Hypothesis with JEPAs* trains text models on pairs of captions from the same image and finds that a JEPA aligns well with a vision encoder, whereas next-token prediction (NTP) given the same caption pairs barely improves over single-caption training. They conclude that the gain comes from the JEPA objective rather than the paired data. However, that baseline is evaluated by mean-pooling the caption's hidden states. Using the hidden state at `[SEP]` (**h([SEP])**, the boundary token between the two captions in training) instead of mean-pooling, the same checkpoint matches the good-view JEPA on Flickr R@1 through one rotation (24.5% vs 24.6%) and closes a lot of the gap in other evaluations. Other readout choices, such as sampling and averaging generated captions, can also improve the alignment significantly, as previous research has already shown for frozen LLMs. Comparisons between NTP and JEPA representations are thus sensitive to how the representation is extracted. A large part of the sample-efficiency gain of JEPA over NTP in the paper may come from the good-view data, not the objective.

In *Proving and Improving the Platonic Representation Hypothesis with JEPAs* [1], Randall Balestriero and Yann LeCun prove under idealised assumptions that JEPA encoders learn the same representation up to one rotation when trained on *good views* (distinct observations of the same scene, such as two captions of one image). *Bad views* corrupt a single observation, such as masking one caption twice; because those views share surface wording, the encoder retains wording alongside scene content. Empirically, a small BERT trained as a good-view JEPA on Flickr captions aligns with frozen DINOv2 much better than a bad-view JEPA or next-token prediction (NTP). The authors argue that NTP acts like a bad-view JEPA whose positives are a prefix and its continuation, and therefore aligns with vision mainly through scale.

I ran a few quick experiments with the help of AI to test that comparison. The paper's paired-caption NTP baseline appears to underestimate what NTP learns from good views because its mean-pooled readout dilutes the cross-caption signal.

1 The paired baseline

The paper's main NTP baseline trains on single captions, which matches the data given to the bad-view JEPA (both score ~.12–.13 against DINOv2 on Flickr) but not the good-view JEPA, which sees which captions share an image. Section 3.1 addresses this with a paired NTP control:

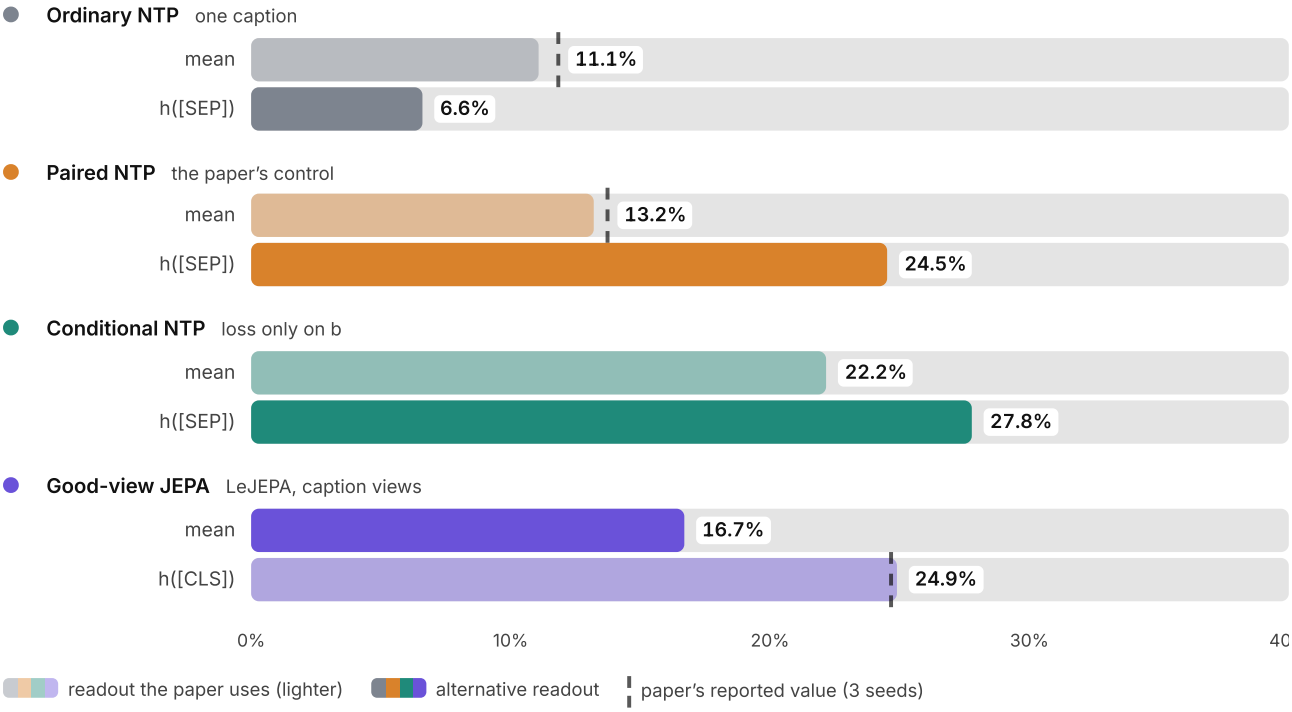
> *"Third, the gain comes from the objective and not from the information that two captions describe the same image: next-token prediction trained on two captions of the same image, [CLS] a [SEP] b [SEP], barely moves (mutual 10-NN .042 to .043, R@1 .021 to .023), i.e., the same positives help next-token prediction far less than they help as JEPA views, as per Proposition 4"*

(Table 1 measures alignment against an image JEPA trained from scratch on Flickr. Against frozen DINOv2 in Table 2, R@1 moves from .118 to .137. All results below use frozen DINOv2.)

The main reason they observe no gain is probably the way they read out the representation of the NTP model. At evaluation time, the paper feeds a single caption `[CLS] a [SEP]` and mean-pools the final hidden states across all tokens. During training, the tokens of *b* attend to all of *a*, so the earlier-layer states of *a* are trained to encode information that helps predict *b*. The final layer is different: each final hidden state is used only to predict the next token at its own position. For the tokens of *a*, that is the next word of *a* (or `[SEP]`). Only the state at `[SEP]`, which has seen all of *a*, is trained to predict the start of *b*. Mean pooling averages mostly over the former. That makes **h([SEP])** an interesting readout candidate.

Appendix B of the paper even tests a last-token (that is, `[SEP]`) readout, but only on single-caption NTP, where it trails mean pooling (R@1 .061 vs .119). That is expected: there `[SEP]` is followed by ignored padding and receives no loss. In the paired model, `[SEP]` predicts the first token of *b*, so it should contain much more information, yet wasn't evaluated separately. My results show that simply using that **h([SEP])** instead of mean pooling already closes much of the gap to the good-view JEPA.

Figure 1 · Same checkpoints, different readout



Flickr30k, BERT-S (33M) text encoders trained from scratch for 200 epochs (one seed each), 1,000 test images against frozen DINOv2 ViT-B/14. R@1 is image→caption retrieval through one orthogonal map; chance is 0.1%. The paired-NTP checkpoint was trained with the authors' code, unchanged.

Using the authors' training code unchanged (BERT-S, Flickr30k, 200 epochs, seed 0) and changing only the readout:

- **R@1, 1k-pair rotation:** 13.2% → 24.5% (good-view JEPA: 24.6%).
- **R@1, 30k-pair rotation:** 23.9% → 32.1% (good-view JEPA: 34.6%).
- **Mutual 10-NN:** .127 → .170 (good-view JEPA: .232).

On retrieval through an orthogonal map (the paper's "direct test of equality up to rotation"), paired NTP read at h([SEP]) matches the good-view JEPA on the 1k-pair fit. The JEPA still clearly leads on mutual 10-NN, but there too the gap closes substantially compared to the mean-pooled readout.

Reading h([SEP]) hurts both single-caption NTP (11.1% → 6.6%) and the JEPA (11.1% vs 24.9% for h([CLS])). It only helps when `[SEP]` precedes a second caption of the *same* image during training: in a 25-epoch pilot where caption *a* was paired with a caption from a different image, h([SEP]) dropped to near chance, as one would expect (Appendix A1).

Other readout choices can help as well. Wang et al. average the hidden states over all layers instead of reading only the last one [5]. On these Flickr models, that raises mean-pooled paired NTP from 13.2% to 16.3% R@1 (1k fit) and ordinary NTP from 11.1% to 14.3%, but both stay far below h([SEP]) (24.5%), and averaging h([SEP]) itself over layers lowers it to 22.2% (Appendix A1). Generic readouts like this help a little across models. For a model trained on pairs (a, b) that describe the same scene, reading the position trained to predict b from all of a targets what the pairing teaches, and is more effective.

In later work, Wang et al. let the model generate a continuation and pool the hidden states over the generated tokens, averaging over several sampled generations [7]. Here, that means sampling four continuations after `[CLS] a [SEP]`, taking the mean of the states that predicted each one's tokens, and averaging the four vectors. For paired NTP on Flickr this changes little: it stays below h([SEP]) on retrieval (22.3% vs 24.5%), though it raises mutual 10-NN (.198 vs .170). At web scale it helps more (below).

2 Predicting only the other caption

Paired NTP still spends roughly half its loss predicting caption a, whereas the JEPA only penalises disagreement between views. To isolate cross-caption prediction, I trained a **conditional** NTP variant using the same `[CLS] a [SEP] b [SEP]` input, initialization, and schedule, but masking the loss on a so only b is predicted (with gradients still flowing into a and both caption orders sampled).

Read at h([SEP]), conditional NTP reaches 27.8% / 38.3% R@1 (1k / 30k fit), ahead of the good-view JEPA's 24.6% / 34.6%, and even mean pooling reaches 22.2% / 33.5%. Its mutual 10-NN (.185) remains below the JEPA's (.232). Part of the gain over standard paired NTP may simply come from b's tokens receiving roughly twice the weight in the loss.

For conditional NTP, sampling does help on Flickr: averaging over four sampled continuations raises it to 28.5% / 41.6% R@1 and its mutual 10-NN to .235, level with the JEPA (.232). The JEPA cannot generate, and the nearest equivalent, averaging the embeddings of several training-style views (crops and masks) of the caption, does not help it: one clean pass stays best (Appendix A5).

Readout choice affects the JEPA as well: its native h([CLS]) readout (the projector input) beats mean pooling on Flickr R@1 (24.9% vs 16.7%), while the projector output itself scores 26.7% on the 1k fit (2.2 points above paired NTP's h([SEP])) and 32.4% on the 30k fit (Appendix A1). On Flickr, the paper thus compares the JEPA at its trained summary token against paired NTP mean-pooled over tokens trained for within-caption prediction. (On WIT-1024 and COCO-5k, the paper instead mean-pools all models across layers following the PRH protocol [2].)

2.1 All numbers

Model · readout	R@1 (1k)	R@1 (30k)	Mutual 10-NN
● Ordinary NTP · mean (paper's readout)	**11.1%**	**19.9%**	**.122**
● Ordinary NTP · h([SEP])	6.6%	12.4%	.049
paper, mean	11.8±0.2	21.3±0.4	.122±.001
● Paired NTP · mean (paper's readout)	13.2%	23.9%	.127
● Paired NTP · h([SEP])	**24.5%**	**32.1%**	.170
● Paired NTP · 4 sampled continuations	22.3%	30.1%	**.198**
paper, mean	13.7±0.4	24.7±1.6	.132±.002
● Conditional NTP · mean (paper's readout)	22.2%	33.5%	.165
● Conditional NTP · h([SEP])	27.8%	38.3%	.185
● Conditional NTP · 4 sampled continuations	**28.5%**	**41.6%**	**.235**
● Good-view JEPA · mean	16.7%	28.5%	.120
● Good-view JEPA · h([CLS]) (paper's readout)	**24.9%**	**33.4%**	**.232**
paper, h([CLS])	24.6±1.5	34.6±0.7	.232±.006
paper, bad-view JEPA	13.3±0.7	20.7±0.8	.132±.001
paper, ordinary NTP early-stopped	13.4±0.8	–	.124±.002

Bold marks the best readout per model. Paper values are three-seed means ± half-range; ours are single seeds.

2.2 Without a fitted map

I also checked three of the paper's evaluations that fit no map: caption→caption retrieval, STS-B sentence similarity, and the Table 6 "scene share" estimator (variance explained by the shared image rather than caption wording). At h([SEP]), paired NTP closes most of the gap to the JEPA on all three (caption retrieval 36.7% → 53.8% vs 60.6%), and conditional NTP slightly exceeds the JEPA on each. These three metrics are text-only; on mutual 10-NN against DINOv2, the JEPA still leads.

Model · readout	Caption→caption R@1	STS-B	Scene share	Mutual 10-NN
● Paired · mean	36.7%	.447	.633	.127
● Paired · h([SEP])	53.8%	**.582**	.791	.170
● Conditional · h([SEP])	**62.3%**	.572	**.816**	.185
● JEPA · h([CLS])	60.6%	.561	.803	**.232**

Single seeds, 200 epochs. More tasks and readouts in Appendix A1.

2.3 Against the paper's image JEPA

The paper's Figure 5 tracks alignment with a separately trained Flickr image JEPA during training and finds that next-token prediction aligns markedly worse than the good-view JEPA (mutual 10-NN .042 vs .062 at the end). Rerunning it with the image JEPA retrained from the authors' code reproduces both endpoints. Read at h([SEP]), paired NTP (.055) and conditional NTP (.058) end most of the way to the JEPA.

Figure 2 · Alignment with the image JEPA during training

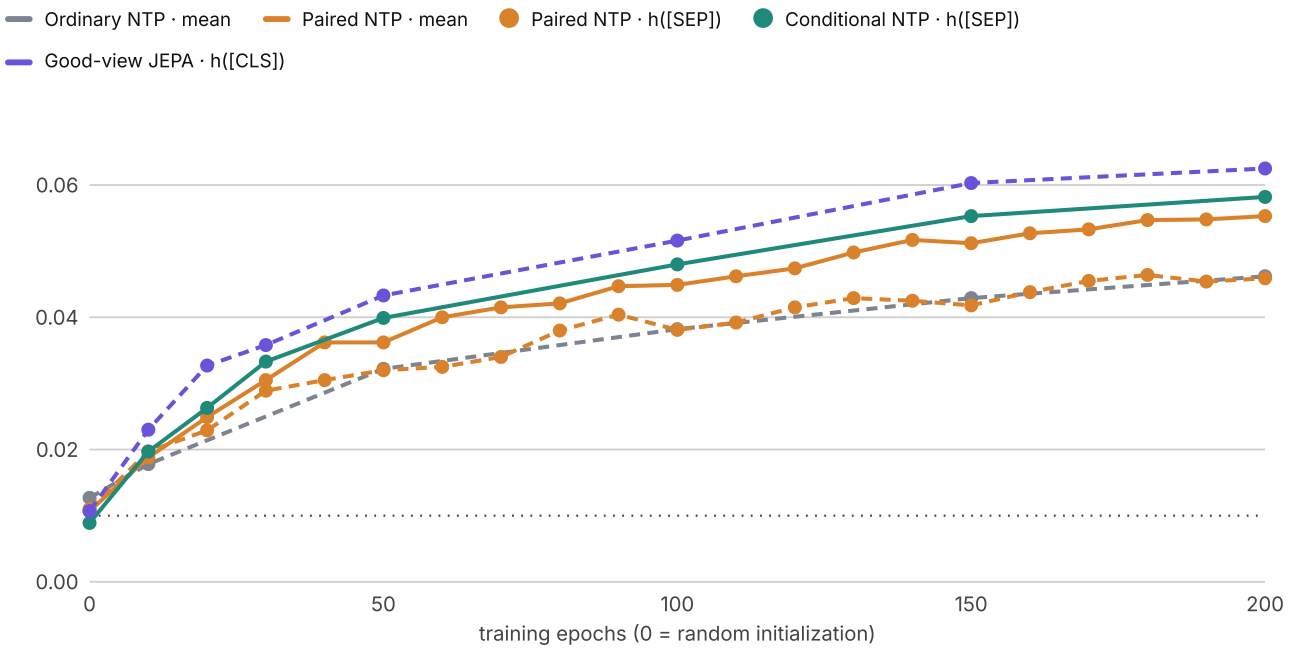


Image side: the paper's Flickr image JEPA (ViT-S/16, 200 epochs, seed 0), retrained with the authors' code, at the same epoch as each text checkpoint. Text side: first caption of the 2,014 val+test pairs, final layer; rotation fitted on the 1,014 val pairs, scored on the 1,000 test pairs. Dashed lines are the paper's readouts, solid lines the alternative. Paired NTP: every 10 epochs from its run's validation dumps; other runs: saved checkpoints. One seed each, so R@1 (a few percent here) is noisy. Dotted line: chance.

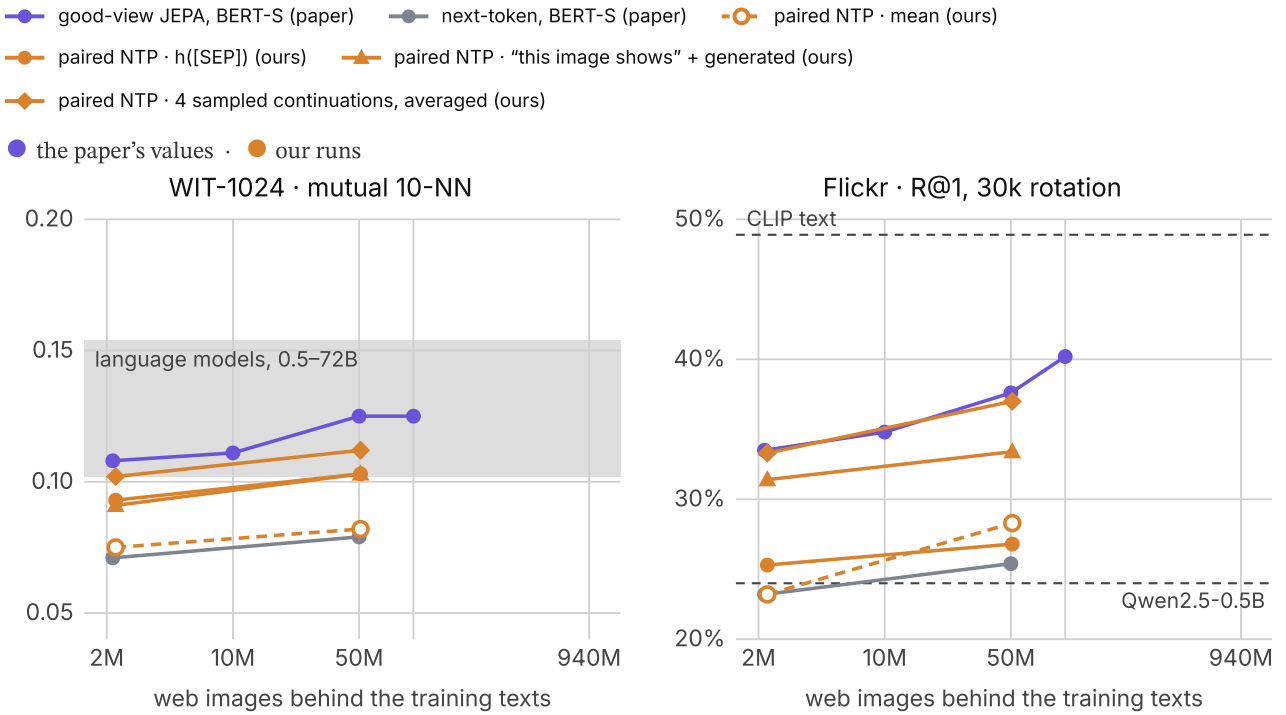
3 At web scale: the 2M and 50M Recap points

To test beyond Flickr, I trained paired BERT-S NTP on two web-scale points in the paper's scaling plot: 2M and 50M. Here, at first glance, the gap still closes, but not as much. But when you look at the data, you can see that a lot of the captions start with something like "The image shows", which could act similarly to the separator token. Reading one token further by appending "the" (which starts nearly every Recap recaption) already closes a lot of the gap. Appending "this image shows" (or "the image shows", which works nearly as well), generating up to 32 tokens and

pooling the hidden states that predicted them also improves the alignment, mainly on Flickr. Averaging four sampled continuations instead of one greedy one, as described above, helps further: paired NTP then comes within 0.6 points of the JEPA on Flickr retrieval at both scales, while still trailing it on WIT and mutual 10-NN.

Model · readout	WIT 10-NN	Flickr R@1 (30k)	Flickr 10-NN
Texts of 2.24M images (20 × 1M samples)			
paper, ordinary NTP (3 seeds)	.070–.071	22.3–24.6%	–
● Paired NTP · mean	.075	23.2%	.103
● Paired NTP · h([SEP])	.093	25.3%	.140
● Paired NTP · “the” (exploratory)	.104	30.1%	.182
● Paired NTP · “this image shows” + generated, pooled (exploratory)	.091	31.4%	.174
● Paired NTP · “this image shows” + 4 sampled continuations, averaged	.102	33.3%	.188
paper, good-view JEPA (3 seeds)	.107 (.095–.115)	33.9%	–
Texts of 50.9M images (40 × 1M samples)			
paper, ordinary NTP	.079	≈25%	–
● Paired NTP · mean	.082	28.3%	.110
● Paired NTP · h([SEP])	.103	26.8%	.129
● Paired NTP · “the”	.114	32.2%	.168
● Paired NTP · “this image shows” + generated, pooled	.103	33.4%	.178
● Paired NTP · “this image shows” + 4 sampled continuations, averaged	.112	37.0%	.197
paper, good-view JEPA (3 seeds)	.125	37.6%	.219

Figure 3 · Paired NTP in the paper’s scaling plot



Text encoders trained on the texts of Recap-DataComp images, never on Flickr or WIT, scored against frozen DINOv2 ViT-B/14: WIT-1024 mutual 10-NN with the PRH protocol (best layer pair) and Flickr image→caption R@1 through one rotation fitted on 30k pairs. Paper values (grey shading, purple, grey and red) are three-seed means where it reports seeds. The paper’s 2M point uses the texts of 2.16M images, ours 2.24M (Table 7’s texts). Ours (orange): one seed per scale, five readouts of the same checkpoint.

4 Readouts for frozen language models

Pretrained LLM baselines in the paper are also mean-pooled over the raw caption. Appending a short prompt and reading the final token’s hidden state (without generating tokens) substantially changes their alignment scores on

both Flickr and WIT-1024. Prompted last-token embeddings are well known [3–7]; here they simply show that the frozen LLM baselines are likewise sensitive to how the representation is read out. Generating continuations and averaging over several samples [7] did not improve on the prompt's final state for Qwen2.5-0.5B and 1.5B (Appendix A5). Full results and Figure 5 are in Appendix A3.

5 Conclusion

The results suggest that a lot of the reported gain in sample efficiency of JEPA vs. NTP comes from the good-view data and not the objective: read out differently than in the paper, NTP trained on the same good-view pairs closes nearly all of the gap to the good-view JEPA on retrieval and most of it on mutual 10-NN. Changing how the representations are read out from the model can change the representation alignment substantially, as previous research has already shown [3–7]. The paper's paired control is read out in a way that understates what it learns.

Several confounds also remain untested. In the paper, hyperparameters were tuned only for the good-view JEPA, while NTP "use[s] the same recipe and [was] not tuned"; each JEPA sample draws eight views from an image's five captions whereas paired NTP sees two; and the JEPA encoder is bidirectional while NTP is causal. All runs here use a single seed, and the largest training runs of the paper remain to be tested (I will not, as I am GPU-poor). I also haven't gone through the theory yet.

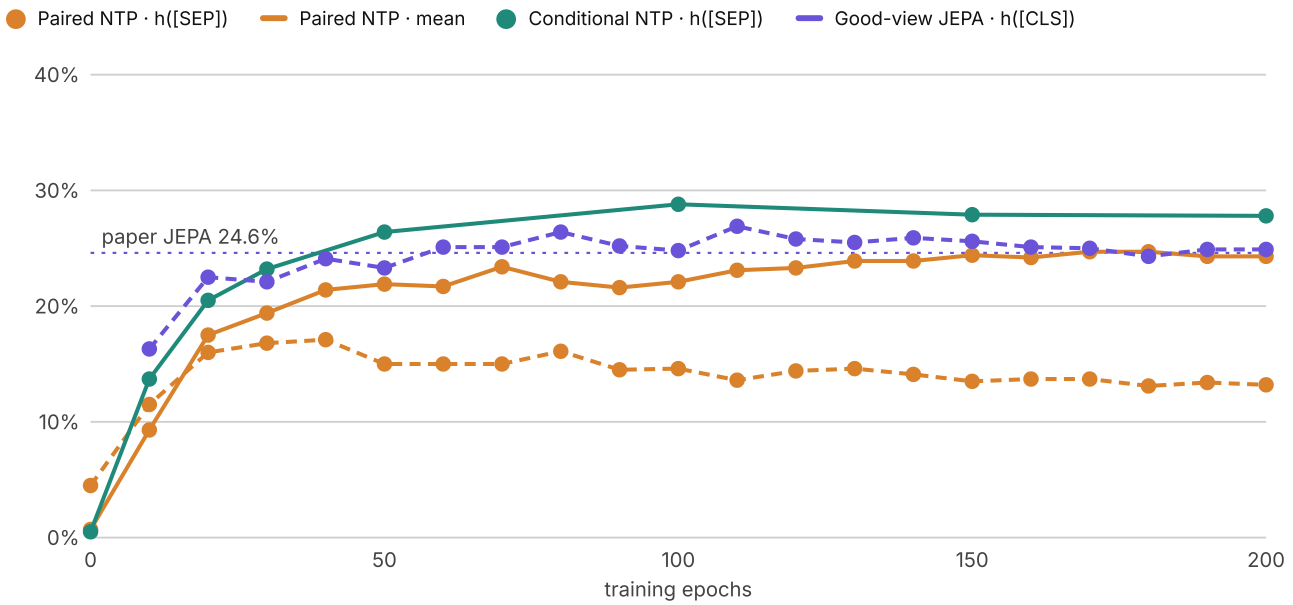
To reproduce the main Flickr result with the authors' repository, train with `train.py --modality text --objective ntp --caption-views` and extract features with `evaluate.py embed <checkpoint> <out> --last-token` (where the last token of `[CLS] a [SEP]` is `[SEP]`).

Disclosure. LLMs from OpenAI, Anthropic and Google wrote and reviewed the code, and helped edit this post. Two independently written pipelines reproduce the paper's baselines (ordinary NTP 11.1% vs reported 11.8%, paired NTP 13.2% vs 13.7%, good-view JEPA 24.9% vs 24.6%, Qwen2.5-0.5B WIT .102 vs .102) and agree on the h([SEP]) effect.

6 Additional results

A1 · More Flickr results

Figure 4 · Alignment during training



Solid lines are the alternative readout, dashed lines the readout the paper uses. Paired NTP: every 10 epochs from the run's own validation outputs. Other runs: saved checkpoints. One seed each. Validation loss is single-caption next-token loss on val+test captions (used for the paper's early-stopped row); because single captions have only ignored padding after [SEP], it never scores predicting *b* from *a*.

Other tasks

The same frozen 200-epoch checkpoints and readouts as in Figure 1, on tasks without a fitted image map (except COCO).

Model · readout	Caption→caption R@1	STS-B	Scene share	COCO R@1
● Ordinary · mean	33.1%	.424	.611	3.3%
● Ordinary · h([SEP])	12.6%	.428	.432	1.5%
● Paired · mean	36.7%	.447	.633	3.4%
● Paired · h([SEP])	53.8%	.582	.791	5.5%
● Conditional · mean	50.6%	.550	.734	6.7%
● Conditional · h([SEP])	**62.3%**	**.572**	**.816**	**7.9%**
● JEPA · h([CLS])	60.6%	.561	.803	6.8%
● JEPA · mean	51.3%	.541	.704	5.0%

JEPA readouts

JEPA readout	R@1 (1k)	R@1 (30k)	Mutual 10-NN
h([CLS]) (paper)	24.9%	33.4%	.232
h([CLS]) → projector	26.7%	32.4%	.233
mean	16.7%	28.5%	.120
h([SEP])	11.1%	21.7%	.071

Averaging over layers

Each readout taken from the final layer (as everywhere else in this post) and from the average of all 12 block outputs, as in Wang et al. [5]. Same checkpoints and protocol as Figure 1. Layer averaging helps mean pooling of both NTP models, but hurts the trained summary positions (h([SEP]) of paired and conditional NTP) and leaves the JEPA's h([CLS]) almost unchanged.

Model · readout	R@1 (1k)	R@1 (30k)	Mutual 10-NN
● Ordinary NTP · mean	11.1% → **14.3%**	19.9% → **23.2%**	.123 → **.130**
● Ordinary NTP · h([SEP])	**6.6%** → 4.1%	**12.4%** → 12.0%	**.049** → .027
● Paired NTP · mean	13.2% → **16.3%**	23.9% → **26.8%**	.127 → **.135**
● Paired NTP · h([SEP])	**24.5%** → 22.2%	**32.1%** → 30.5%	**.170** → .158
● Conditional NTP · mean	**22.2%** → 21.1%	33.5% → **34.5%**	.165 → **.173**
● Conditional NTP · h([SEP])	**27.8%** → 23.9%	**38.3%** → 36.9%	**.185** → .146
● Good-view JEPA · h([CLS])	**24.9%** → 24.5%	**33.4%** → 32.0%	.232 → **.236**
● Good-view JEPA · mean	**16.7%** → 16.6%	**28.5%** → 26.4%	.120 → **.123**

Control: caption pairs from different images

To check whether h([SEP]) benefits simply from having any second caption after `[SEP]`, a 25-epoch pilot included a shuffled control. It matches conditional NTP (loss only on b), except caption a comes from a different image in the batch. With unrelated context, the model ignores a: single-caption embeddings collapse to effective rank ~1, and alignment drops to near chance under both readouts. These are 25-epoch pilot checkpoints, not the 200-epoch runs; in those, h([SEP]) only overtakes mean pooling after about 20–30 epochs (Figure 4).

Pilot arm (25 epochs) · readout	R@1 (1k)	R@1 (30k)	Mutual 10-NN	Eff. rank
● Paired · mean	18.1%	26.5%	.145	40
● Paired · h([SEP])	18.9%	27.1%	.160	18
● Conditional · mean	20.2%	30.0%	.162	32
● Conditional · h([SEP])	**22.9%**	**31.1%**	.160	18
● Shuffled · mean	1.3%	2.3%	.027	1.1
● Shuffled · h([SEP])	1.2%	3.4%	.020	1.5

Pre-declared check: appending "a"

To test whether reading one token into caption *b* also helps on Flickr, I appended "a" (the most common first word in Flickr training captions, chosen before scoring) to give `[CLS] a-caption [SEP] a`. Paired NTP is read at the appended "a", while the JEPA is still read at h([CLS]) as a control for adding the token (reading the JEPA at "a" performed much worse). Evaluated on the 200-epoch Figure 1 checkpoints: paired NTP improves on 30k-pair retrieval (+2.1), mutual 10-NN, and caption retrieval, is flat on 1k-pair retrieval, and dips on STS-B; the JEPA control gains +0.9 on 30k-pair retrieval and a little on STS-B. One seed; not significant on its own.

Metric	NTP · h([SEP])	NTP · "a"	JEPA · h([CLS])	JEPA · h([CLS]) + "a"
Flickr R@1, 1k fit	24.5%	24.3%	24.9%	24.9%
Flickr R@1, 30k fit	32.2%	34.3%	33.4%	34.3%
Mutual 10-NN	.170	.185	.232	.232
Caption→caption R@1	53.8%	55.3%	60.6%	61.1%
STS-B Spearman	.582	.574	.561	.572
Scene share (k = 32)	.791	.795	.803	.804
COCO R@1 (Flickr means)	5.4%	5.7%	6.8%	6.9%

A2 · Web-scale readout search (exploratory)

All readouts of the Recap paired-NTP checkpoint (final, seed 0) that were tried, reported equally. Prefixes are appended after `[SEP]` as the beginning of a caption *b*; the state of the last prefix token is read. Caption *a* is unchanged and nothing is generated. These were scored on the same Flickr and WIT test sets, so the best entry is optimistically selected.

One untested reading of why "the" beats h([SEP]) here: in Recap, caption *b* is either an alt-text (short, varied) or a LLaVA recaption ("The image shows..."). So part of what h([SEP]) is trained to predict is which kind of caption comes next, which is about format rather than the scene. Appending "the" settles that, and the state can encode content instead. On Flickr all captions are of one kind, which would fit appending "a" helping only a little there (Appendix A1). The final-layer state at "the" itself mostly predicts "image", but like h([SEP]) it gets its scene content from the lower layers, where all of *b*'s later tokens attend to it.

Readout	Flickr R@1 (1k)	Flickr R@1 (30k)	Flickr 10-NN	WIT 10-NN
full mean (paper)	16.4%	23.2%	.103	.075
h([SEP])	16.5%	25.3%	.140	.093
last caption token	10.9%	19.7%	.056	.049
caption-only mean	15.4%	22.4%	.096	.072
mean without [SEP]	15.5%	22.5%	.096	.072
"this image shows"	14.8%	23.0%	.091	.102
"the image shows"	15.7%	22.5%	.087	.102
"a photo of"	15.6%	23.9%	.111	.093
"this text says"	15.7%	24.1%	.153	.062
"the"	**21.2%**	**30.1%**	**.182**	**.104**

Generating a continuation and pooling its hidden states (8 or 32 greedy tokens, with or without a prefix) reaches at most 31.4% Flickr R@1 (30k; "this image shows" + 32 tokens) and .095 on WIT: no clear gain over appending "the", and worse on WIT. Wang et al. report that pooling generated tokens beats single tokens for pretrained LLMs [7]; with one greedy continuation this small trained model doesn't show that, but averaging several sampled ones does (Appendix A5).

A3 · Frozen Qwen2.5: prompted readouts and all 15 conditions

Appending a short prompt and reading the final token's hidden state (without generating tokens) substantially changes the frozen Qwen2.5 baselines:

```
Caption: {A}
Alternative caption of the same scene:
```

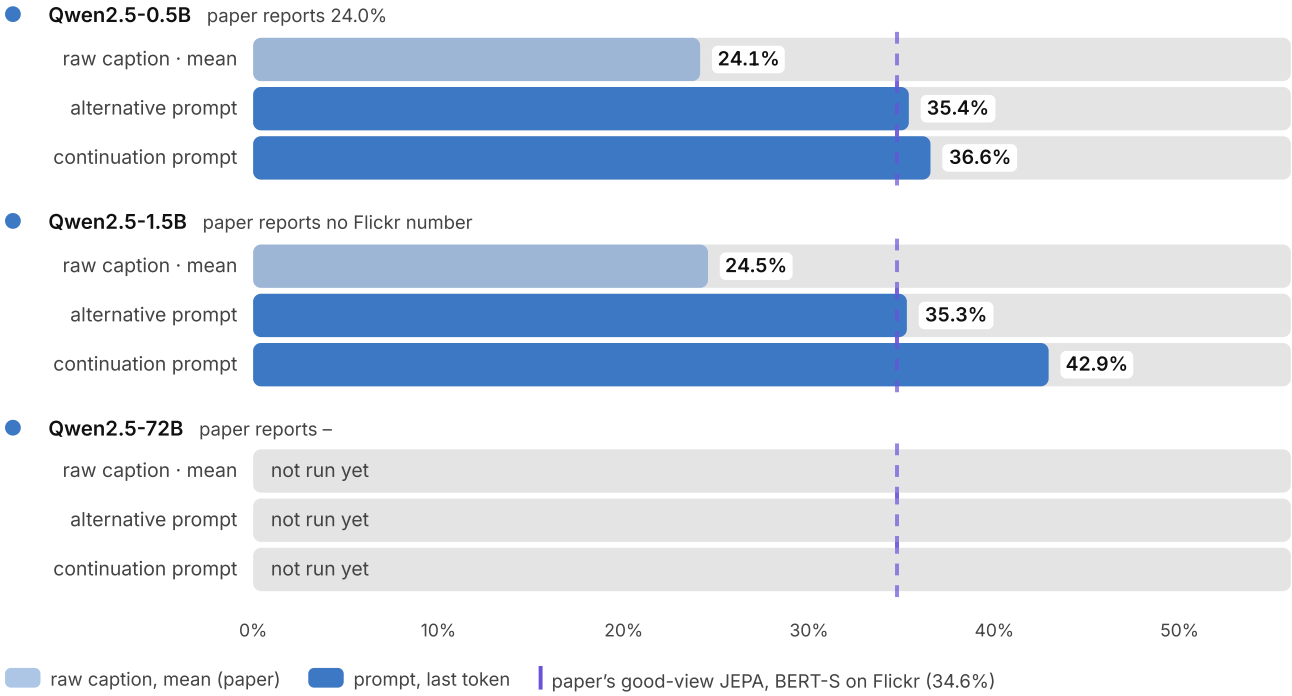
Under the paper's Flickr protocol (final layer, 30k-pair rotation), Qwen2.5-0.5B R@1 jumps from **24.1% to 35.4%** (14.0% to 22.2% with a 1k-pair fit), passing the Flickr-trained good-view JEPA (34.6%). On Qwen2.5-1.5B, a continuation prompt reaches **42.9%** (raw caption 24.5%, alternative prompt 35.3%). On WIT-1024, mutual 10-NN rises from **.102 to .138** (0.5B), **.108 to .155** (1.5B),

and **.153 to .170** (72B). The relative gain shrinks with scale, the continuation prompt hurts at 72B, and 72B remains below the paper's 109M web-trained JEPA (.184).

Prompt wording itself may matter less than providing a post-caption summary position: "Continuation of this caption:" matches "Alternative caption of the same scene:" on 0.5B and beats it on 1.5B Flickr, whereas bare markers like literal "[SEP]" or Qwen's end-of-text token do not help.

Closest to this, Wang et al. raise frozen Qwen3's alignment with DINOv2 on WIT with sensory prompts ("imagine what it would be like to see…") [5]. They let the model generate a continuation and average its hidden states over the generated tokens and over all layers; in a single forward pass averaged over the input tokens, the prompt did not help. The prompted last-token state here needs no generation. Wang et al. also find that mean-pooling over generated tokens beats single-token readouts [7]. Here, greedily generating an alternative caption and pooling its states did not beat the single prompted last-token state (33.9% vs 35.4% Flickr R@1 on 0.5B), and averaging over all layers, which Wang et al. find often beats any single layer [5], does not change that (33.6% vs 34.5%). On the frozen models, layer averaging gives mixed results: on 1.5B it lifts the alternative prompt to 39.2% Flickr R@1 but lowers the continuation prompt from 42.9% to 40.2%, and on the raw caption it falls from 24.1% to 12.4% (0.5B) unless each layer is normalised before averaging (26.9%).

Figure 5 · Frozen Qwen2.5: raw caption vs prompted readout



Flickr image→caption R@1 through one orthogonal map, final hidden layer (no layer search), 1,000 test images against frozen DINOv2 ViT-B/14. This is the paper's Flickr protocol for pretrained models; it reports .135 / .240 for Qwen2.5-0.5B with the raw caption. 0.5B and 1.5B have been run (1.5B in FP16). The prompts were chosen after looking at results, so read these as exploratory.

All 15 readout conditions

WIT columns use the PRH protocol (best layer pair). The Flickr column uses the paper's Flickr protocol: final layer, rotation fitted on 30k pairs, R@1 among 1,000 test captions.

Condition	0.5B · WIT	1.5B · WIT	72B · WIT	0.5B · Flickr R@1	1.5B · Flickr R@1
`raw_mean`	.102	.108	.153	24.1%	24.5%
`raw_last`	.073	.077	.091	20.2%	–
`alternative`	.138	.155	.170	35.4%	35.3%
`continuation`	.139	.150	.146	36.6%	42.9%
`echo`	.113	.128	.155	36.7%	–
`cls_sep_mean`	.098	.107	.108	24.3%	–
`cls_sep_last`	.108	.118	.121	23.1%	–
`cls_sep_echo`	.118	.130	.126	32.3%	–
`native_eos_mean`	.098	.105	.149	23.7%	–
`native_eos_last`	.051	.072	.067	1.6%	–
`generated_alternative`	.103	.126	.133	33.9%	–
`isolated_alternative`	.104	.119	.136	27.8%	–
`generated_continuation`	.113	.119	.138	27.5%	–
`isolated_continuation`	.107	.110	.131	22.2%	–
`cls_sep_generated_mean`	.114	.130	.131	33.0%	–

A4 · Methods, reproduction and caveats

- **Metrics.** Mutual 10-NN and R@1 follow the authors' `prh.metrics.alignment`. The 1,014 validation pairs (or 30,014 train+validation pairs) fit the means and one orthogonal map. Scores are on the 1,000 Flickr test pairs, using one caption per image.
- **Readouts.** All are from the final layer, except the layer averages in Appendices A1 and A3. "Mean" averages all non-padding positions, including [CLS] and [SEP]. h([SEP]) is the state at the final [SEP] of `[CLS] a [SEP]`. Evaluation inputs never contain a second caption.
- **Uncertainty.** Every training arm has one seed. The paper's seed spread for paired NTP is ±0.4 (1k) and ±1.6 (30k) points. With 1,000 test queries, the standard error of R@1 near 25% is about 1.4 points. The h([SEP]) gain for paired NTP (+11 points) is far outside both, but the smaller differences between conditional NTP and JEPA are not.
- **Selection.** The Flickr test set was looked at repeatedly while exploring readouts. h([SEP]) was proposed beforehand, for the mechanistic reason above, but the prompt variants for frozen LLMs were not.
- The JEPA was trained with the authors' code and default recipe. The seed-0 run collapsed around epoch 98, when its h([CLS]) became nearly constant and R@1 fell to 0.2%. The seed-1 run shown here stayed healthy and matches the paper (24.9% vs 24.6%, mutual 10-NN .232 vs .232).
- The paired NTP run was also trained with the authors' unchanged code (prh-proof `a2b9949`, stable-pretraining `ab836bf`, PyTorch 2.11). The conditional run used the same pinned code, with only the loss mask changed.
- The ordinary (single-caption) NTP run was trained with a reimplementation of the paper's recipe, not the authors' code. It reproduces the paper's baseline (11.1% vs 11.8% R@1, mutual 10-NN .122 vs .122).
- The web-scale runs use the repository's `--data recaptext:11` (2.24M images, 20 epochs of 1M samples) and `--data recaptext:250` (50.9M images, 40 epochs), with `--objective ntp --caption-views --lr 1e-4 --clip 1.0`.
- Flickr-trained models of every kind barely transfer to WIT-1024 (mutual 10-NN ≈ .03), as the paper also notes, so I don't use WIT for them.

A5 · Sampled continuations, view averaging and real captions

After `[CLS] a [SEP]` (plus "this image shows" for the Recap models), four continuations of up to 32 tokens are sampled at temperature 0.7 and top-p 0.95. Each is pooled as the mean of the states that predicted its tokens, and the four vectors are

averaged, as Wang et al. do with several sampled generations [7]. This one setting was fixed before scoring, with one sampling seed. Ordinary NTP ends every continuation at once, so it is left out.

Model	greedy	1 sample	2 samples	4 samples	4 vs greedy
Recap, paired NTP, prefix "this image shows"					
2M	31.4% · .174	28.9% · .167	32.2% · .186	33.3% · .188	+1.9 [−0.5, +4.3]
50M	33.4% · .178	32.0% · .175	36.5% · .190	37.0% · .197	+3.6 [+1.3, +5.9]
Flickr, 200 epochs, no prefix					
paired NTP	25.3% · .164	20.8% · .159	26.5% · .188	30.1% · .198	+4.8 [+2.0, +7.5]
conditional NTP	39.3% · .220	36.0% · .213	38.0% · .230	41.6% · .235	+2.3 [+0.0, +4.8]

A single sample is worse than greedy decoding; the gain comes from averaging. Letting the 50M model generate 64 greedy tokens instead of 32 does not help (32.4% vs 33.4%), and pooling only the middle or the later tokens hurts (24.7%, 23.8%).

The JEPA. It cannot generate, but it was trained on views of a caption: spans keeping 50–100% or 15–50% of the tokens, with 15% masked. Averaging h([CLS]) over K such views of the test caption never beats one clean pass (33.4%): masking alone comes closest (32.7% at $K = 8$), crops reach 32.4%, the training mix of views 26.6%, and adding crops to the clean pass lowers it to about 30%.

Real captions as a ceiling. Each test image has four more human captions. Averaging over all five raises the JEPA to 52.5% R@1 (mutual 10-NN .310) and conditional NTP at h([SEP]) to 54.6% (.251). Used as continuations in place of the samples, four real captions take conditional NTP to 49.6%, against 41.6% with four samples. Given the same captions, both models align about equally well, and the JEPA gains as much from extra captions as NTP does (+19 vs +16 points).

Frozen Qwen2.5. With the alternative-caption prompt (Appendix A3), generating does not beat reading the prompt's final state: four samples give 21.2% vs 22.1% R@1 (1k fit) for 0.5B and 24.0% vs 23.4% for 1.5B, and lower WIT scores (.123 vs .138, .132 vs .154). Pooling 128 generated tokens instead of 32 hurts.

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```bibtex
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